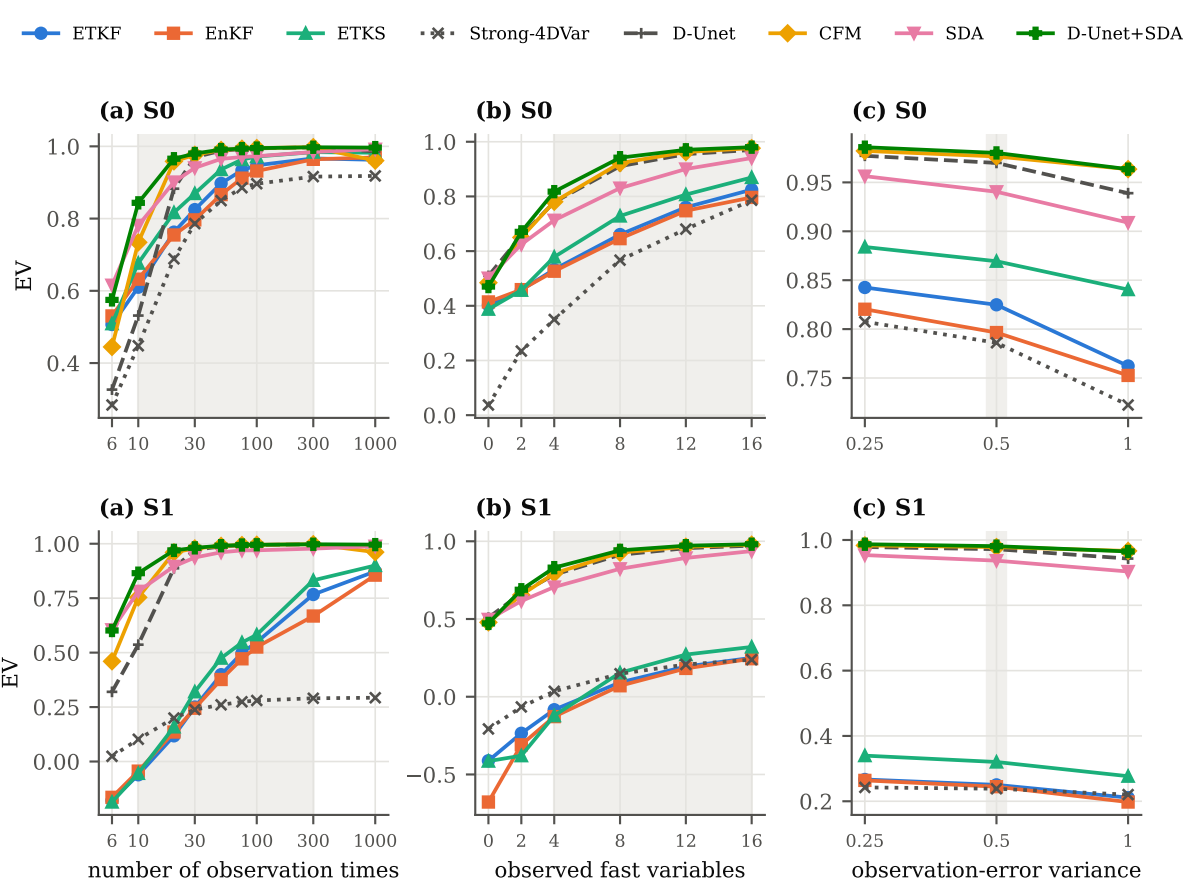
ETKF
EnKF
ETKS
Strong-4DVar
D-Unet
CFM
SDA
D-Unet+SDA
(a) S0
(b) S0
(c) S0
(a) S1
(b) S1
(c) S1
EV
EV
number of observation times
observed fast variables
observation-error variance